

Distinction Bytes (Mobile App)

Product Requirements Document (PRD)

1. Product Overview

Product Name: Distinction Bytes

Platform: Mobile (Android & iOS)

Primary Audience: Learners (secondary school, tertiary, professional, lifelong learners)

One-liner:

Distinction Bytes is a gamified, AI-powered micro-learning mobile app that enables learners to study any subject or topic through bite-sized, personalized content—either as part of structured courses on Distinction or through on-demand topic exploration.

2. Problem Statement

Current learning platforms suffer from one or more of the following:

- Long, exhausting content not optimized for mobile
- Poor learner engagement and retention
- Lack of personalization
- Fragmented tools (LMS, videos, quizzes, notes)
- Weak motivation systems

Learners want:

- Short, engaging lessons
 - Clear progress and rewards
 - Ability to learn *anything*, anytime
 - Guidance without rigidity
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3. Product Goals & Success Metrics

3.1 Goals

- Create a **daily learning habit** for users
- Improve course completion rates on Distinction
- Enable instant learning from curiosity-driven searches
- Leverage FlexiSAF AI for adaptive, personalized learning

3.2 Success Metrics (North-Star & Supporting)

- **Daily Active Learners (DAL)**
 - 7-day and 30-day retention
 - Average bytes completed per session
 - Streak continuation rate
 - Course completion rate (for enrolled learners)
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4. Target Users & Personas

Persona 1: University Student

- Needs exam-focused learning
- Short attention windows
- Values progress tracking

Persona 2: Skill Learner / Professional

- Goal-oriented
- Wants clarity & relevance
- Interested in certificates

Persona 3: Curious Learner

- Exploratory
 - Searches topics often
 - Needs structure from chaos
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5. Core Value Proposition

Distinction Bytes uniquely combines:

- **Micro-learning** (3–7 minute bytes)
 - **AI-generated & curated content**
 - **Gamification that rewards consistency**
 - **Structured courses + open exploration**
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6. Key Product Principles

1. Learner-first, not institution-first
 2. Mobile-native, not LMS-adapted
 3. Motivation over pressure
 4. Progress over perfection
 5. AI as a silent guide, not a lecturer
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7. Core Features & Requirements

7.1 Onboarding & User Setup

Functional Requirements:

- Email / phone / SSO login (reuse Distinction auth)
- Select learning goals
- Choose interests & level
- Optional placement assessment

Non-Functional:

- < 60 seconds to complete onboarding
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7.2 Learning Modes

A. Course Mode

For enrolled Distinction learners

Requirements:

- Sync enrolled courses via Distinction API
- Display course → modules → bytes

- Daily learning goals
- Progress visualization

B. Explore Mode

For open learning

Requirements:

- Topic search input
 - AI generates a mini-learning path
 - Bytes ordered by difficulty
 - Optional save to library
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7.3 Learning Byte

Definition:

A byte is the smallest unit of learning.

Components:

- Title
- Core explanation (text + optional audio)
- Visual aid (diagram / card)
- 1–3 quick questions

Requirements:

- Must load in < 1s
 - Offline caching (premium)
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7.4 Gamification System

Mechanics:

- XP per byte
- Daily streaks
- Topic mastery levels
- Badges

Rules:

- Reward consistency more than speed

- No punishment for breaks
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7.5 Assessments & Feedback

- Quick quizzes
 - Adaptive difficulty
 - Instant explanations
 - Mastery thresholds
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7.6 AI-Powered Personalization (FlexiSAF-AI)

Capabilities:

- Generate bytes from topic prompts
- Adjust difficulty
- Recommend next bytes
- Multiple explanation styles

Constraints:

- AI output must follow byte templates
 - Human-verified content for flagship courses
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8. Technical Architecture

8.1 Frontend

- React Native
- Shared design system
- Offline support

8.2 Backend

Existing Systems:

- Distinction API (courses, users, progress)
- FlexiSAF-AI service (content generation, personalization)

New Services Needed:

- Byte orchestration service
 - Gamification service
 - Recommendation engine (AI-assisted)
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9. Data & Analytics

Track:

- Session length
- Bytes completed
- Drop-off points
- Streaks
- Search queries

Use GA4 + internal analytics

10. Security & Compliance

- Token-based auth (reuse Distinction)
 - Rate limiting on AI endpoints
 - Content moderation layer
 - GDPR-aligned data handling
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11. Monetization Strategy (Phase-Based)

Phase 1

- Free access
- Focus on engagement

Phase 2

- Premium subscription
 - Institution-sponsored access
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12. MVP Scope (Phase 1)

Include:

- Auth & onboarding
- Course mode
- Explore mode
- Learning bytes
- Basic gamification

Exclude:

- Social features
 - Live classes
 - Certificates
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13. Roadmap (High-Level)

0–3 Months: MVP build & internal beta

3–6 Months: Public launch, AI refinement

6–12 Months: Monetization & scale

14. Open Questions

- Offline limits for free users?
 - AI content approval workflow?
 - Institutional branding support?
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Owner: FlexiSAF / Distinction Product Team

Status: Draft PRD v1.0

15. Structure of Bytes

Executive Recommendation (TL;DR)

**Default to Text-first + Visual Cards,
Enhance with Optional Audio,
Use Video very selectively,
and let AI adapt the format per learner and per topic.**

In short:

Bytes should be *readable in 30–90 seconds*, understandable without sound, and delightful to look at.

1. The “Byte” Is Not a Mini Lecture

This is the biggest trap to avoid.

A byte is:

-  NOT a shortened video lecture
-  NOT a PDF page
-  NOT a slide deck

A byte is:

A single idea, explained clearly, with just enough depth to move the learner forward.

If a concept needs more than one idea → it becomes **multiple bytes**.

2. Recommended Default Byte Structure (This Is the Core)

Every byte should follow the same **mental rhythm**, even if the media differs.

The 5-Part Byte Model

1. Hook (1–2 lines)

- Why should I care?

2. Core Explanation (Text)

- Clear, conversational, concise

3. Visual Anchor

- Diagram, card, illustration, or table

4. Quick Check

- 1–3 questions or tap-to-reveal

5. Takeaway

- One sentence summary

This structure works across *any subject*.

3. Text: Your Primary Medium (Yes, Really)

Text should be the **foundation**, not an afterthought.

Why text wins:

- Fast to consume
- Works offline
- Easy to scan
- Lowest cognitive load
- Cheapest to generate & refine
- Best for AI adaptation

But not “wall of text”

Think:

- Short paragraphs
- Bullet points
- Highlighted keywords
- Cards instead of pages

 **Rule:** If a byte takes more than ~90 seconds to read → split it.

4. Visuals: The Memory Anchor

Visuals are what make bytes **stick**.

What kind of visuals?

- Simple diagrams
- Flow arrows
- Comparison tables
- Illustrated examples
- Concept cards

What to avoid:

- Stock photos
- Decorative images
- Heavy infographics

Think:

“If the learner screenshots this, will it still make sense?”

If yes → good visual.

5. Audio: Optional but Powerful

Audio should be **additive**, not mandatory.

How audio should work:

- Tap  to “listen to this byte”
- 30–90 seconds max
- Natural, calm voice
- Mirrors text (not extra content)

Why this matters:

- Commute learning
- Accessibility
- Low bandwidth markets
- Fatigue reduction

Later phase:

- Multiple audio styles:
 - Calm tutor
 - Energetic coach
 - Exam-focused narrator

6. Video: Use Sparingly (Very Sparingly)

Video is expensive and often unnecessary.

When video makes sense:

- Demonstrations (e.g., lab experiment, forex chart reading)
- Motion-based concepts
- Skill walkthroughs

When it doesn't:

- Definitions
- Theoretical explanations
- Concept breakdowns



Rule:

If a concept can be understood without motion → don't use video.

7. Interactive Micro-Interactions (Underrated Weapon)

Instead of video, use **interaction**.

Examples:

- Tap to reveal answer
- Drag-and-drop matching
- Swipe to compare
- Highlight the correct part of a diagram

These:

- Increase retention
- Feel game-like
- Cost less than video
- Work offline

8. AI-Driven Format Adaptation (Where You Win)

This is where Distinction Bytes can leapfrog competitors.

Same byte, different presentations:

- “Explain simply”
- “Exam-focused”
- “Practical example”
- “Analogy-based”

Also:

- Weak learner → more visuals, slower pace
- Advanced learner → concise text, fewer prompts

The **content stays the same**, the **presentation adapts**.

9. Byte Format Types (Standardize These)

Don't allow unlimited creativity early. Define **formats**.

Suggested MVP formats:

1. **Explain** – concept breakdown
2. **Compare** – A vs B
3. **Step-by-step** – processes
4. **Example** – worked problems
5. **Practice** – questions only

This helps:

- AI consistency
 - Analytics
 - Content QA
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10. Accessibility & Inclusion (Important for Scale)

Design bytes so that:

- They work without sound
- They work on small screens
- They load fast on low bandwidth
- They're readable in dark mode

Africa + global markets demand this.

Final Advice (If I Had to Be Opinionated)

If Distinction Bytes were my product, I'd say:

Text + visual cards as default
Audio as a tap-away enhancement
Video only when motion is essential
Interaction over passive consumption
AI adapts the presentation, not the learner

This approach is:

- Scalable

- Cost-effective
- Delightful
- Future-proof

If you want, next we can:

- Define the **Byte JSON schema**
- Design **AI prompt templates for each byte type**
- Create **3–5 sample bytes** (e.g., Calculus, Forex, Biology)
- Or map **content standards for Distinction vs open exploration**

11. Byte Sources (How Bytes are Created)

A) Course-Generated Bytes (Course Mode)

Purpose: Help learners study enrolled Distinction courses through AI-generated micro-learning bytes.

User Flow:

1. User logs in and sees enrolled courses
2. User selects a course
3. User selects module/topic or chooses “Generate from whole course”
4. System generates a learning path of bytes and begins the Byte Player

Inputs to Byte Generation:

- Enrolled course content (course outline, modules, topics, learning objectives)
- User profile (level, learning goals, interests)
- Optional user constraints (exam prep, revision, simplified explanations)

Requirements:

- Fetch enrolled courses via Distinction API
 - Allow course selection and optional module/topic selection
 - Generate a structured byte path (e.g. 5–30 bytes depending on course/module size)
 - Order bytes by progression: beginner → intermediate → advanced
 - Allow regeneration if user is not satisfied (limited retries)
 - Store generated bytes for the user (cache/persist) to reduce repeated generation costs
 - Track progress per generated byte path and reflect in course progress
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B) User-Generated Bytes (Manual Topic Mode)

Purpose: Let learners instantly learn anything by providing a topic and context.

User Inputs:

- Topic (required)
- Level (Beginner / Intermediate / Advanced)
- Goal (exam prep, revision, skill-building)
- Explanation style (simple, examples-first, deep)
- Optional: number of bytes to generate (e.g., 3–10)

Requirements:

- Generate a mini learning path from the user topic
- Order bytes by difficulty progression
- Allow saving generated bytes/paths to library (optional in MVP)
- Ensure AI follows the byte template strictly

12. Bytes data structure

Topic → Byte Path → Bytes

Topic

└ BytePath

└ Byte 1 (concept intro)

└ Byte 2 (rule)

└ Byte 3 (example)

└ Byte 4 (application)

This applies to:

- Manual Topic Mode
- Course Mode (course or module becomes the “topic source”)